International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B103: Network Forensics Fundamentals

LAB 9: WEP40 Wireless Packet Decryption and Aircrack Forensics

Three-Week Delivery Block 3/3 | 1-2 August 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction

Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.

Course Code	CIP-B103
Course Title	Network Forensics Fundamentals
Lab Number	Lab 9
Lab Title	WEP40 Wireless Packet Decryption and Aircrack Forensics
Student Name	Fuseini Imoru Kantuogaa
Student ID	17291
Instructor Name	Dr. Sarah James
Delivery Block	3/3 of 3
Scheduled Dates	1-2 August 2026
Estimated Duration	4-5 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	Historical CodeGate CTF file.xz/file capture supplied for offline analysis

1. Source Materials and Slide Alignment

- Decrypt Wireless Packets: IEEE 802.11 frames, WEP/RC4, IV reuse, Aircrack-ng key recovery, airdecap-ng decryption, file carving and optional passphrase analysis.

- The slide uses the historical CodeGate CTF 2015 Good Crypto capture and the known WEP40 key A4:3D:F6:F3:74.
- The assignment asks for host/server IP addresses, MAC addresses, manually extracted images/HTML and the router password where the optional CTF challenge is completed.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded CIP-B103 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A historical wireless CTF capture protected by deprecated WEP40 has been supplied. You will preserve and decompress the evidence, identify IEEE 802.11 traffic, recover or validate the 40-bit key using Aircrack-ng, decrypt the capture offline, inventory network endpoints, carve transferred files and explain why WEP is cryptographically unsafe.

3. Learning Outcomes

- Identify management, control and data frames in an 802.11 capture.
- Explain WEP40 structure, RC4 keystream use, IV reuse and integrity limitations.
- Use Aircrack-ng on a supplied historical capture only.
- Decrypt WEP traffic using airdecap-ng and verify output files.
- Extract IP/MAC endpoints, protocols, images and HTML from decrypted traffic.
- Distinguish a WEP key from a human passphrase and document optional CTF password analysis.
- Recommend modern wireless security controls.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.

- Do not capture, crack or connect to any live wireless network. This lab is exclusively offline and limited to the supplied historical training file.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux VM	Offline capture analysis.
Wireless tools	aircrack-ng suite	Recover/validate key and decrypt capture.
Packet tools	Wireshark and tshark	Inspect encrypted/decrypted frames.
Carving tools	foremost, binwalk, file, strings	Recover transferred objects.
Compression	xz-utils	Decompress file.xz.
Evidence	Historical CodeGate CTF capture	Authorized offline dataset.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/CIP-B103-Lab9/{evidence,working,exported,reports,screenshots,scripts}
cd ~/CIP-B103-Lab9
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```
sudo apt update
sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
wget -O evidence/file.xz \
  'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
file working/file_working
sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
```

Screenshot required: Compressed evidence, decompressed capture type and SHA-256 hashes.

```
┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/CIP-B103-Lab9/{evidence,working,exported,reports,screenshots,scripts}

┌──(kali㉿kali)-[~]
└─$ cd ~/CIP-B103-Lab9

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ pwd
/home/kali/CIP-B103-Lab9

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find . -maxdepth 1 -type d -print
.
./working
./screenshots
./evidence
./scripts
./reports
./exported
```

Screenshot 1: Evidence folder structure created (mkdir, cd, pwd, find).

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
[sudo] password for kali:
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
foremost is already the newest version (1.5.7-11+b2).
binwalk is already the newest version (2.4.3+dfsg1-2).
The following packages were automatically installed and are no longer required:
  libraw23t64  libvidstab1.1  openjdk-21-jre  openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.

Upgrading:
  aircrack-ng  liblzma-dev  liblzma5  xz-utils

Summary:
  Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 1713
  Download size: 1,297 kB / 1,998 kB
  Space needed: 59.4 kB / 23.4 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 xz-utils amd64 5.8.3-1 [740 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 aircrack-ng amd64 1:1.7+git20230807.4
```

Screenshot 2: aircrack-ng, wireshark, tshark, foremost, binwalk, xz-utils installed/upgraded.

```
Get:2 http://http.kali.org/kali kali-rolling/main amd64 aircrack-ng amd64 1:1.7+git20230807.4
bf83f1a-3 [557 kB]
Fetched 1,297 kB in 3s (445 kB/s)
(Reading database… 432542 files and directories currently installed.)
Preparing to unpack …/liblzma-dev_5.8.3-1_amd64.deb…
Unpacking liblzma-dev:amd64 (5.8.3-1) over (5.8.2-2)…
Preparing to unpack …/liblzma5_5.8.3-1_amd64.deb…
Unpacking liblzma5:amd64 (5.8.3-1) over (5.8.2-2)…
Setting up liblzma5:amd64 (5.8.3-1)…
(Reading database… 432542 files and directories currently installed.)
Preparing to unpack …/xz-utils_5.8.3-1_amd64.deb…
Unpacking xz-utils (5.8.3-1) over (5.8.2-2)…
Preparing to unpack …/aircrack-ng_1%3a1.7+git20230807.4bf83f1a-3_amd64.deb…
Unpacking aircrack-ng (1:1.7+git20230807.4bf83f1a-3) over (1:1.7+git20230807.4bf83f1a-2+b1)…
Setting up aircrack-ng (1:1.7+git20230807.4bf83f1a-3)…
Setting up xz-utils (5.8.3-1)…
Setting up liblzma-dev:amd64 (5.8.3-1)…
Processing triggers for libc-bin (2.42-13)…
Processing triggers for man-db (2.13.1-1)…
Processing triggers for kali-menu (2026.1.5)…
```

Screenshot 3: Package installation continued.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ wget -O evidence/file.xz 'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/co
degate-ctf-2015/programming/good-crypto/file.xz'
--2026-08-11 18:18:00--  https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegat
e-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.1
10.133, 185.199.111.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... c
onnected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: ‘evidence/file.xz’

evidence/file.xz        100%[═══════════════════════════════⇒]  12.24M  1.34MB/s    in 9.4s

2026-08-11 18:18:11 (1.31 MB/s) - ‘evidence/file.xz’ saved [12837660/12837660]
```

Screenshot 4: file.xz downloaded via wget from the CodeGate CTF 2015 write-ups repository.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ ls -lh evidence/
total 13M
-rw-rw-r-- 1 kali kali 13M Aug 11 18:18 file.xz

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  evidence/file.xz
```

Screenshot 5: Evidence file listed and SHA-256 hashed.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ cp --preserve=timestamps evidence/file.xz working/file_working.xz

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ unxz -k working/file_working.xz

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ file working/file_working
working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11
, capture length 65535)

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_working
```

Screenshot 6: Working copy created, decompressed with unxz, identified as an 802.11 pcap, and hashed.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tree
.
├── evidence
│   └── file.xz
├── exported
├── reports
│   ├── file_xz_sha256.txt
│   └── working_hashes.txt
├── screenshots
├── scripts
└── working
    ├── file_working
    └── file_working.xz

7 directories, 5 files

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$
```

Screenshot 7: Evidence folder tree after decompression.

7. Mini Evidence and Chain-of-Custody Worksheet

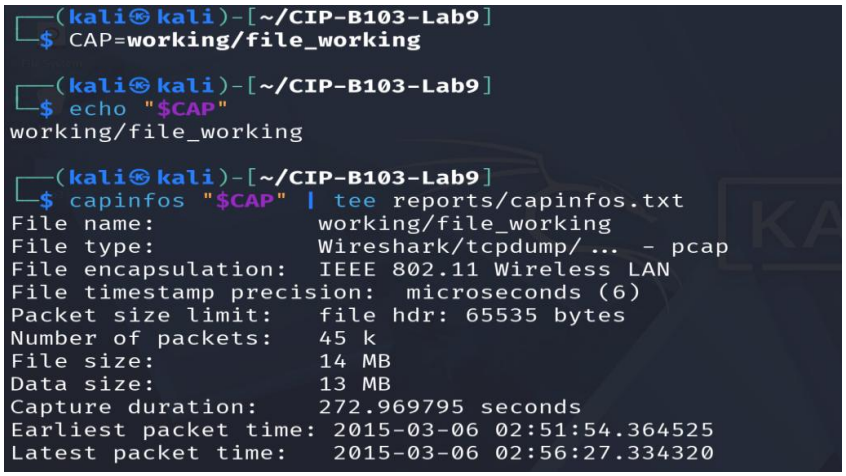
Field	Student Entry
Case/lab identifier	CIP-B103-Lab9-Fuseini Imoru Kantuogaa
Trainee name	Fuseini Imoru Kantuogaa
Date and time started	Tue 11 August 2026, approx. 18:18 EDT (Kali VM) — timestamp of the wget download of the evidence file
Evidence file name(s)	evidence/file.xz (working copy: working/file_working.xz, decompressed to working/file_working, decrypted to working/file_working-dec)

Field	Student Entry
Source or generation method	Downloaded via wget from the historical CodeGate CTF 2015 'Good Crypto' write-ups GitHub repository — a public historical training capture, not locally generated
Original SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 (evidence/file.xz)
Working-copy SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 (working/file_working.xz, identical to original); decompressed working/file_working: c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
Analysis workstation/VM	Kali Linux (user: kali), CIP-B103-Lab9 working directory
Notes on any changes	cp --preserve=timestamps used for the working copy; SHA-256 of evidence and working .xz copies matched exactly. The WEP40 key A4:3D:F6:F3:74 was independently recovered via aircrack-ng's PTW attack (100% decrypted), confirming rather than merely assuming the historical slide-provided key.

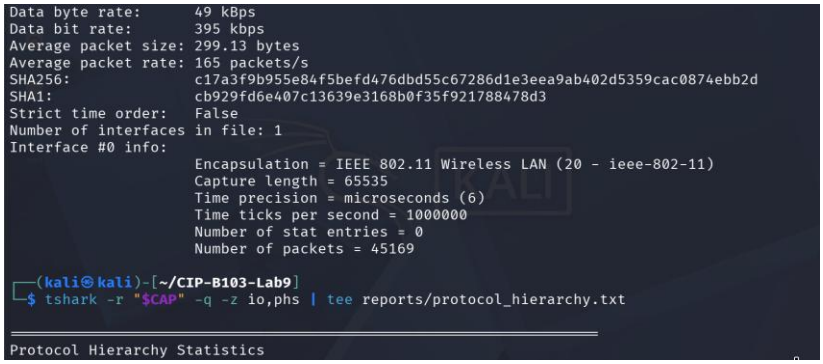
8. Part A - Inventory the Wireless Capture

```
CAP=working/file_working
capinfos "$CAP" | tee reports/capinfos.txt
tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
tshark -r "$CAP" -Y 'wlan' -T fields \
  -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
  | head -n 100 | tee reports/wlan_frame_sample.tsv
```

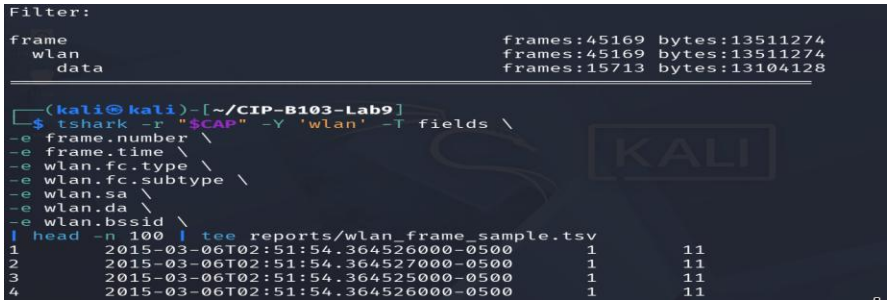
Frame Category	Type Value	Purpose	Example Evidence
Management	0	Association, authentication, beacons	Frame 54 — Beacon (subtype 8) from 00:26:66:55:97:d6 to Broadcast, advertising the ESSID ‘cgnetwork’
Control	1	Flow control and acknowledgements	Frames 1–53 (and thousands more) — subtype 11 (RTS/CTS/Block-Ack family) dominate the early wlan_frame_sample.tsv output
Data	2	Carries network payload	Frame 10130 (and 15,477 similar frames) — WEP-protected data between BSSID 00:26:66:55:97:d6 and stations f0:f6:1c:68:96:7c / 00:26:66:55:97:d4, readable only after airdecap-ng decryption



Screenshot 8: CAP variable set; capinfos showing 45,169 packets over ~273 seconds (802.11 encapsulation).



Screenshot 9: capinfos concluded (SHA256/SHA1); tshark protocol-hierarchy statistics started.



Screenshot 10: Protocol hierarchy (wlan/data) and start of the wlan frame-type/subtype field extraction.

```
5       2015-03-06T02:51:54.365550000-0500      1       11
6       2015-03-06T02:51:54.366063000-0500      1       11
7       2015-03-06T02:51:54.366062000-0500      1       11
8       2015-03-06T02:51:54.366061000-0500      1       11
9       2015-03-06T02:51:54.366063000-0500      1       11
10      2015-03-06T02:51:54.366061000-0500      1       11
11      2015-03-06T02:51:54.366064000-0500      1       11
12      2015-03-06T02:51:54.366063000-0500      1       11
13      2015-03-06T02:51:54.366063000-0500      1       11
14      2015-03-06T02:51:54.366063000-0500      1       11
15      2015-03-06T02:51:54.366064000-0500      1       11
16      2015-03-06T02:51:54.366062000-0500      1       11
17      2015-03-06T02:51:54.366575000-0500      1       11
18      2015-03-06T02:51:54.366575000-0500      1       11
19      2015-03-06T02:51:54.366573000-0500      1       11
20      2015-03-06T02:51:54.366574000-0500      1       11
21      2015-03-06T02:51:54.366575000-0500      1       11
22      2015-03-06T02:51:54.366575000-0500      1       11
23      2015-03-06T02:51:54.366575000-0500      1       11
24      2015-03-06T02:51:54.366574000-0500      1       11
25      2015-03-06T02:51:54.366574000-0500      1       11
```

Screenshot 11: wlan frame sample continued — predominantly control frames (subtype 11).

```
26      2015-03-06T02:51:54.366574000-0500      1       11
27      2015-03-06T02:51:54.366574000-0500      1       11
28      2015-03-06T02:51:54.366574000-0500      1       11
29      2015-03-06T02:51:54.366574000-0500      1       11
30      2015-03-06T02:51:54.366575000-0500      1       11
31      2015-03-06T02:51:54.366573000-0500      1       11
32      2015-03-06T02:51:54.366575000-0500      1       11
33      2015-03-06T02:51:54.366574000-0500      1       11
34      2015-03-06T02:51:54.366575000-0500      1       11
35      2015-03-06T02:51:54.366575000-0500      1       11
36      2015-03-06T02:51:54.366574000-0500      1       11
37      2015-03-06T02:51:54.366575000-0500      1       11
38      2015-03-06T02:51:54.366576000-0500      1       11
39      2015-03-06T02:51:54.366575000-0500      1       11
40      2015-03-06T02:51:54.366574000-0500      1       11
41      2015-03-06T02:51:54.366573000-0500      1       11
42      2015-03-06T02:51:54.366573000-0500      1       11
43      2015-03-06T02:51:54.366574000-0500      1       11
44      2015-03-06T02:51:54.367085000-0500      1       11
45      2015-03-06T02:51:54.367086000-0500      1       11
46      2015-03-06T02:51:54.367087000-0500      1       11
```

Screenshot 12: wlan frame sample continued.

```
47      2015-03-06T02:51:54.367089000-0500      1       13
48      2015-03-06T02:51:54.367085000-0500      1       11
49      2015-03-06T02:51:54.367087000-0500      1       11
50      2015-03-06T02:51:54.367087000-0500      1       11
51      2015-03-06T02:51:54.367087000-0500      1       11
52      2015-03-06T02:51:54.367087000-0500      1       11
53      2015-03-06T02:51:54.367086000-0500      1       11
54      2015-03-06T02:51:54.367633000-0500      0       8       00:26:66:55:97:d6       ff:ff
:ff:ff:ff:ff    00:26:66:55:97:d6
55      2015-03-06T02:51:54.368111000-0500      1       11
56      2015-03-06T02:51:54.368111000-0500      1       11
57      2015-03-06T02:51:54.368110000-0500      1       11
58      2015-03-06T02:51:54.368110000-0500      1       11
59      2015-03-06T02:51:54.368110000-0500      1       11
60      2015-03-06T02:51:54.368110000-0500      1       11
61      2015-03-06T02:51:54.368622000-0500      1       11
62      2015-03-06T02:51:54.368623000-0500      1       11
63      2015-03-06T02:51:54.368621000-0500      1       11
64      2015-03-06T02:51:54.368622000-0500      1       11
65      2015-03-06T02:51:54.368621000-0500      1       11
66      2015-03-06T02:51:54.368623000-0500      1       11
```

Screenshot 13: wlan frame sample continued, including the first Beacon frame (frame 54).

```
67      2015-03-06T02:51:54.368622000-0500      1       11
68      2015-03-06T02:51:54.368623000-0500      1       11
69      2015-03-06T02:51:54.368622000-0500      1       11
70      2015-03-06T02:51:54.368622000-0500      1       11
71      2015-03-06T02:51:54.368623000-0500      1       11
72      2015-03-06T02:51:54.368621000-0500      1       11
73      2015-03-06T02:51:54.368622000-0500      1       11
74      2015-03-06T02:51:54.369134000-0500      1       11
75      2015-03-06T02:51:54.369134000-0500      1       11
76      2015-03-06T02:51:54.369135000-0500      1       11
77      2015-03-06T02:51:54.369648000-0500      1       11
78      2015-03-06T02:51:54.369646000-0500      1       11
79      2015-03-06T02:51:54.370159000-0500      1       11
80      2015-03-06T02:51:54.371182000-0500      1       11
81      2015-03-06T02:51:54.371181000-0500      1       11
82      2015-03-06T02:51:54.371183000-0500      1       11
83      2015-03-06T02:51:54.371694000-0500      1       11
84      2015-03-06T02:51:54.371694000-0500      1       11
85      2015-03-06T02:51:54.372207000-0500      1       11
86      2015-03-06T02:51:54.372718000-0500      1       11
87      2015-03-06T02:51:54.375279000-0500      1       11
```

Screenshot 14: wlan frame sample continued.

```
88      2015-03-06T02:51:54.375791000-0500      1       11
89      2015-03-06T02:51:54.376813000-0500      1       11
90      2015-03-06T02:51:54.376815000-0500      1       11
91      2015-03-06T02:51:54.376815000-0500      1       11
92      2015-03-06T02:51:54.376815000-0500      1       11
93      2015-03-06T02:51:54.376814000-0500      1       11
94      2015-03-06T02:51:54.376815000-0500      1       11
95      2015-03-06T02:51:54.376814000-0500      1       11
96      2015-03-06T02:51:54.381421000-0500      1       11
97      2015-03-06T02:51:54.381423000-0500      1       11
98      2015-03-06T02:51:54.382446000-0500      1       11
99      2015-03-06T02:51:54.382446000-0500      1       11
100     2015-03-06T02:51:54.382957000-0500      1       11

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tree
.
├── evidence
│   └── file.xz
├── exported
└── reports
```

Screenshot 15: wlan frame sample concluded (100 frames); evidence folder tree started.

```
└─$ tree
.
├── evidence
│   └── file.xz
├── exported
├── reports
│   ├── capinfos.txt
│   ├── file_xz_sha256.txt
│   ├── protocol_hierarchy.txt
│   ├── wlan_frame_sample.tsv
│   └── working_hashes.txt
├── screenshots
├── scripts
└── working
    ├── file_working
    └── file_working.xz

7 directories, 8 files

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$
```

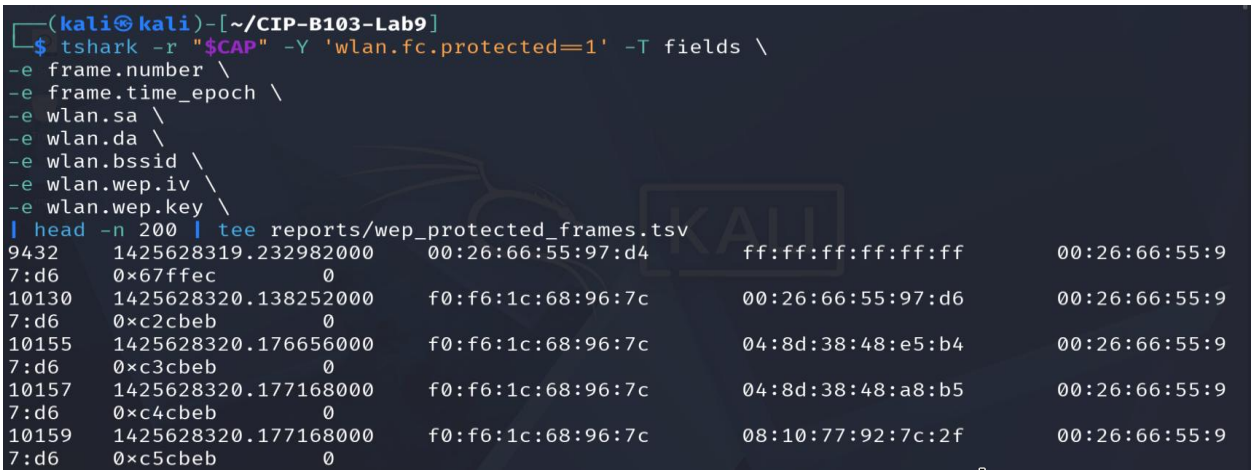
Screenshot 16: Evidence folder tree confirming the new reports files.

9. Part B - Identify WEP Protection and Key Parameters

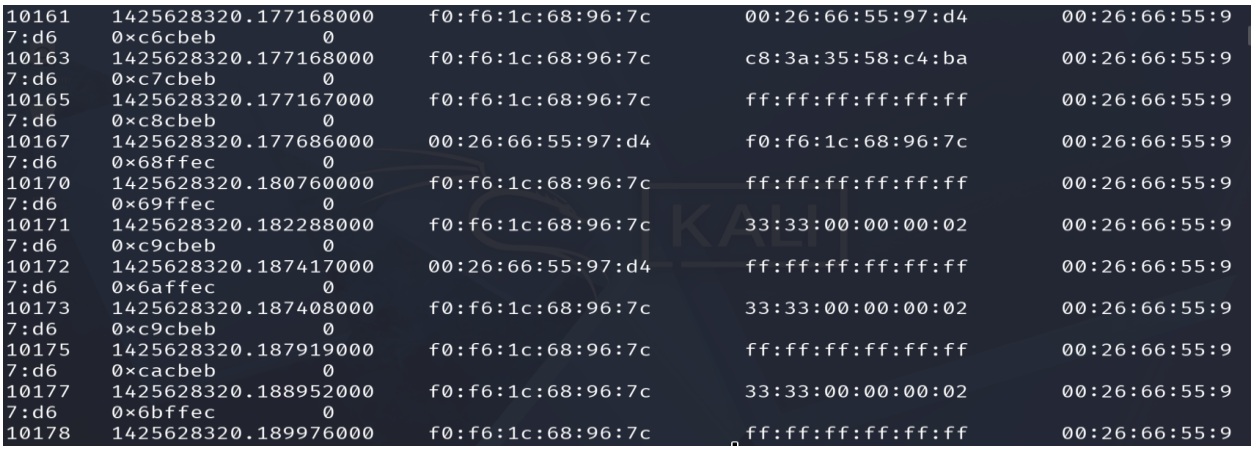
1. Confirm that protected data frames are present.
2. Record BSSID, source/destination MAC addresses and IV-related fields where dissected.
3. Explain that 64-bit WEP normally combines a 24-bit IV with a 40-bit secret key.
4. Explain why repeated IVs and RC4 weaknesses make WEP unsuitable.

```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
  -e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e
wlan.wep.key \
  | head -n 200 | tee reports/wep_protected_frames.tsv

tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
  | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
```



Screenshot 17: WEP-protected frame field extraction started (BSSID, source/destination MAC, WEP IV).



Screenshot 18: Protected frame extraction continued, showing sequential WEP IVs.

```
7:d6    0×6cffec        0
10183   1425628320.192016000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×cbcbeb        0
10185   1425628320.193048000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×6dffec        0
10224   1425628320.259088000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×cccbeb        0
10228   1425628320.262680000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×6effec        0
10229   1425628320.262678000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×6fffec        0
10233   1425628320.263184000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×cdcbeb        0
10261   1425628320.419350000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×cecbeb        0
10266   1425628320.420374000    f0:f6:1c:68:96:7c       01:00:5e:00:00:02       00:26:66:55:9
7:d6    0×cfcbeb        0
10268   1425628320.420374000    f0:f6:1c:68:96:7c       01:00:5e:00:00:fb       00:26:66:55:9
7:d6    0×d0cbeb        0
10270   1425628320.420886000    f0:f6:1c:68:96:7c       01:00:5e:00:00:02       00:26:66:55:9
7:d6    0×70ffec        0
```

Screenshot 19: Protected frame extraction continued.

```
10272   1425628320.422421000    f0:f6:1c:68:96:7c       01:00:5e:00:00:fb       00:26:66:55:9
7:d6    0×71ffec        0
10353   1425628320.583190000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×72ffec        0
10355   1425628320.583700000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×73ffec        0
10357   1425628320.583702000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×d1cbeb        0
10359   1425628320.584726000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×74ffec        0
10360   1425628320.586774000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×d2cbeb        0
10367   1425628320.789014000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×d3cbeb        0
10375   1425628320.936471000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×d4cbeb        0
10377   1425628320.936469000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×75ffec        0
10379   1425628320.937495000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×76ffec        0
10380   1425628320.962071000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
```

Screenshot 20: Protected frame extraction continued.

```
7:d6    0×d5cbeb        0
10382   1425628320.976407000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×d6cbeb        0
10388   1425628321.112150000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×77ffec        0
10390   1425628321.139798000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×d7cbeb        0
10392   1425628321.156181000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×d8cbeb        0
10394   1425628321.157208000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×78ffec        0
10401   1425628321.287763000    f0:f6:1c:68:96:7c       33:33:00:00:00:16       00:26:66:55:9
7:d6    0×d9cbeb        0
10403   1425628321.287766000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×79ffec        0
10405   1425628321.287766000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×7affec        0
10407   1425628321.287767000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×7bffec        0
10409   1425628321.288279000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×7cffec        0
```

Screenshot 21: Protected frame extraction continued.

```
10411   1425628321.289816000    f0:f6:1c:68:96:7c       33:33:00:00:00:16       00:26:66:55:9
7:d6    0×7dffec        0
10412   1425628321.321554000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×dacbeb        0
10414   1425628321.325143000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×7effec        0
10415   1425628321.327698000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×dbcbeb        0
10417   1425628321.327698000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×dccbeb        0
10419   1425628321.328214000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×7fffec        0
10426   1425628321.462355000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×ddcbeb        0
10431   1425628321.488979000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×decbeb        0
10433   1425628321.490008000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×80ffec        0
10446   1425628321.813588000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×81ffec        0
10448   1425628321.841235000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
```

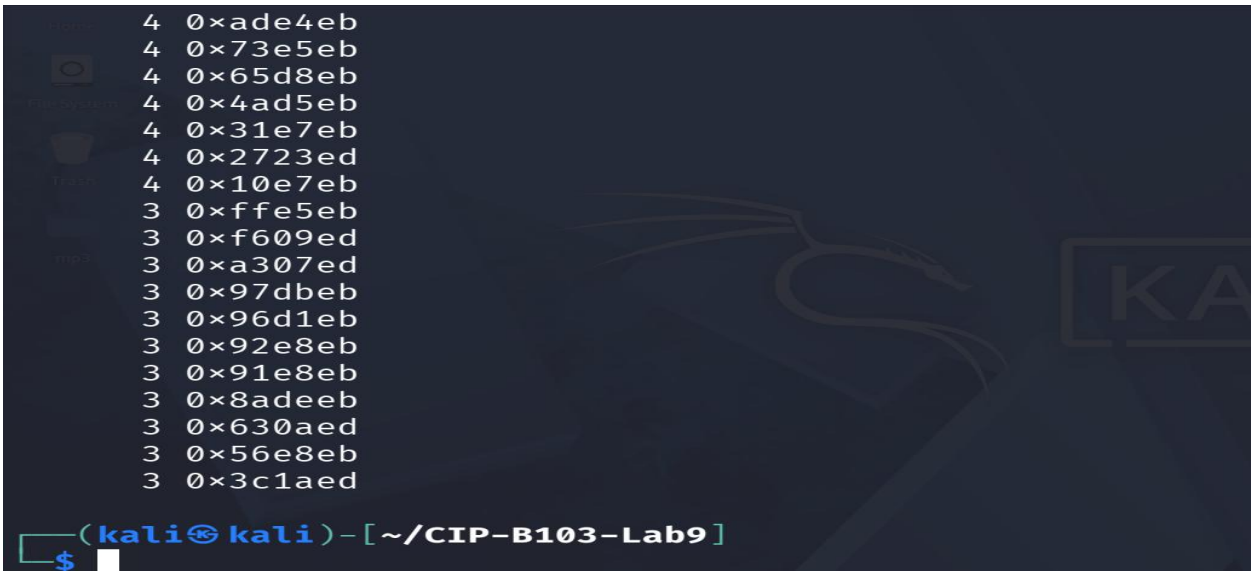
Screenshot 22: Protected frame extraction concluded (200-frame sample); repeated-IV summary started.

```
7:d6    0×dfcbeb        0
10458   1425628322.163860000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×82ffec        0
10460   1425628322.163860000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×83ffec        0
10462   1425628322.193046000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×e0cbeb        0
10464   1425628322.194070000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×e1cbeb        0
10476   1425628322.516627000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×84ffec        0
10478   1425628322.544276000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×e2cbeb        0
10480   1425628322.545302000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×85ffec        0
10481   1425628322.545300000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×86ffec        0
10483   1425628322.546325000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×e3cbeb        0
10485   1425628322.547860000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×e4cbeb        0
```

Screenshot 23: Repeated-IV summary continued, showing IVs recurring multiple times.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
-e wlan.wep.iv \
| sort | uniq -c | sort -nr | head -n 30 \
| tee reports/repeated_iv_summary.txt
      7 0×42e8eb
      7 0×28d3eb
      6 0×95dfeb
      5 0×9415ed
      5 0×86e5eb
      5 0×7ae5eb
      5 0×28e7eb
      5 0×19e5eb
      4 0×d5d7eb
      4 0×d3e8eb
      4 0×c915ed
      4 0×c7d8eb
      4 0×ade4eb
      4 0×73e5eb
      4 0×65d8eb
```

Screenshot 24: Repeated-IV summary command and top results (IVs repeating 5–7 times).



Screenshot 25: Repeated-IV summary concluded.

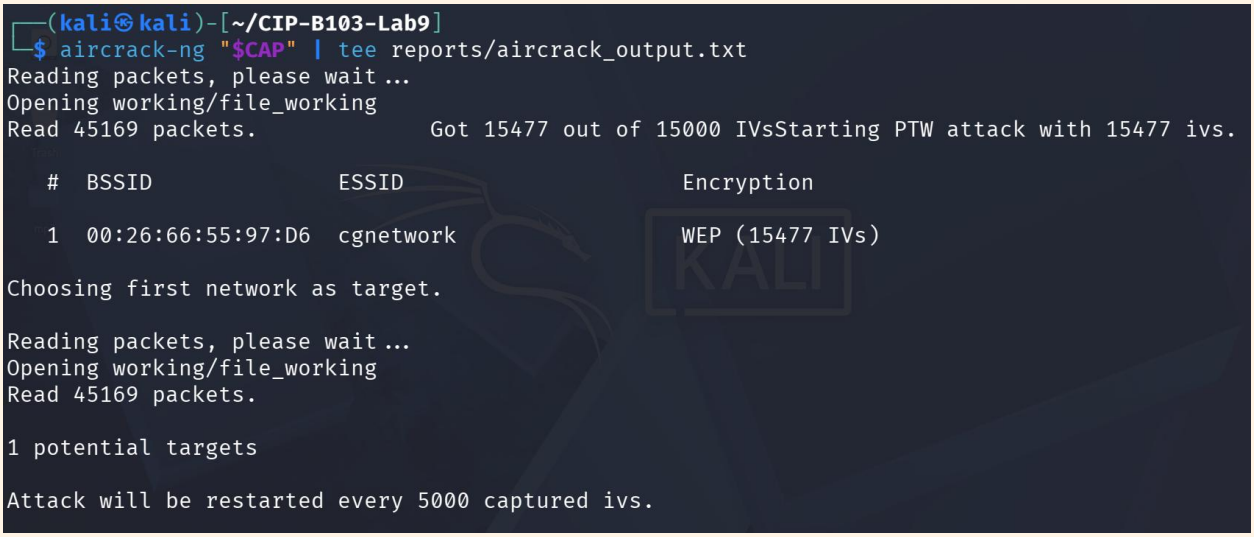
10. Part C - Recover or Validate the WEP40 Key

Run Aircrack-ng against the supplied capture. Record selected network/BSSID, number of IVs, key length and recovered key. The slide-provided key is A4:3D:F6:F3:74, entered without colons for airdecap-ng.

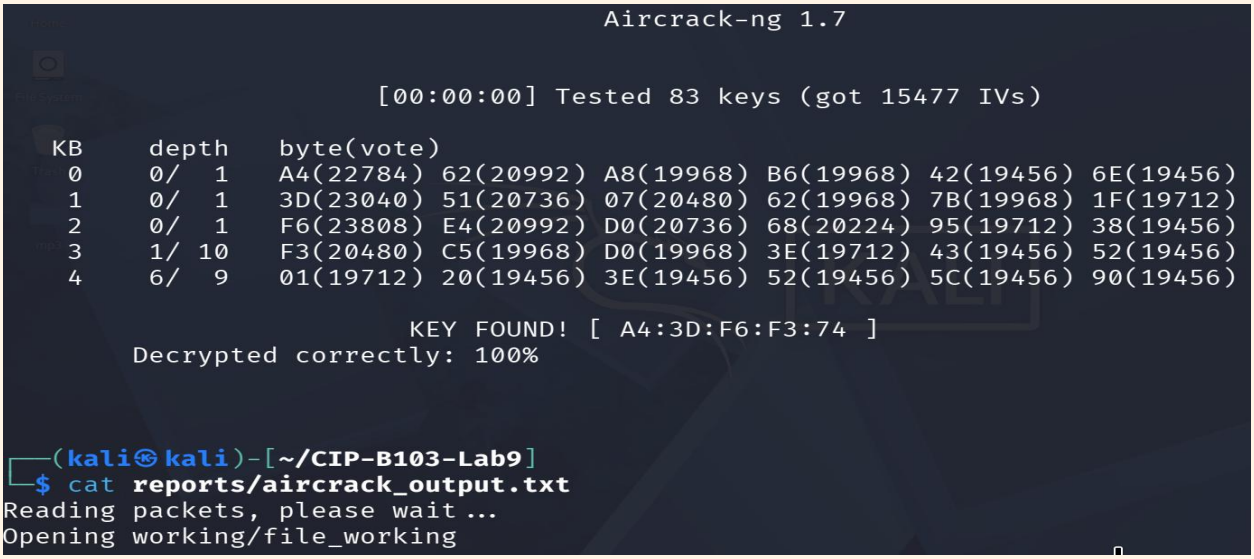
```
aircrack-ng "$CAP" | tee reports/aircrack_output.txt
# Expected historical training key from the slide: A4:3D:F6:F3:74
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
```

Evidence Handling

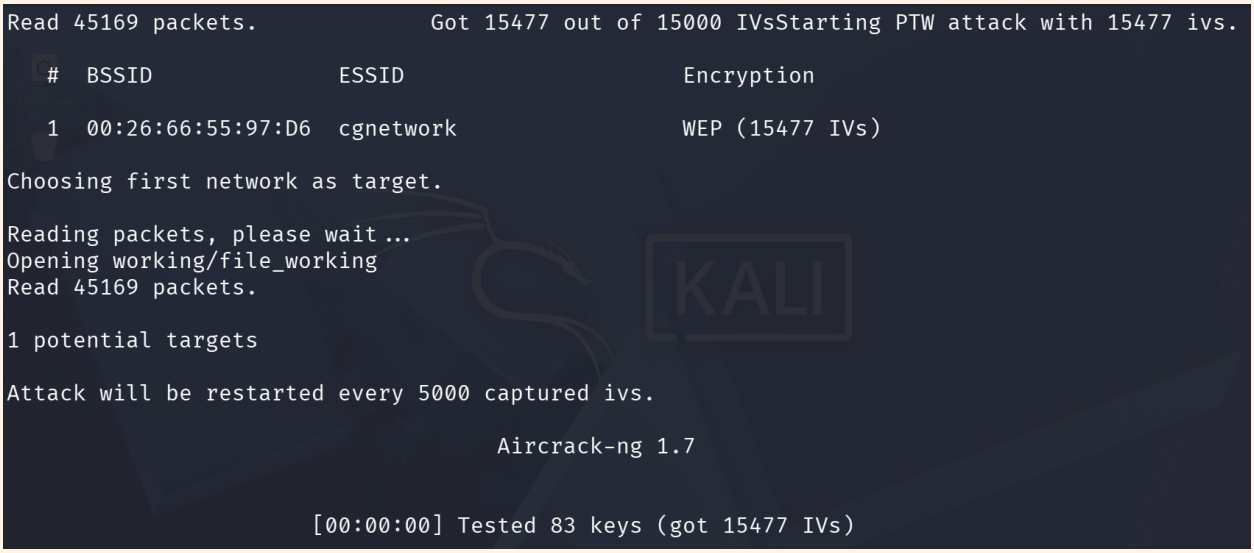
The recovered key is part of the historical CTF dataset, not authorization to attack any wireless network. Keep it inside the course submission.



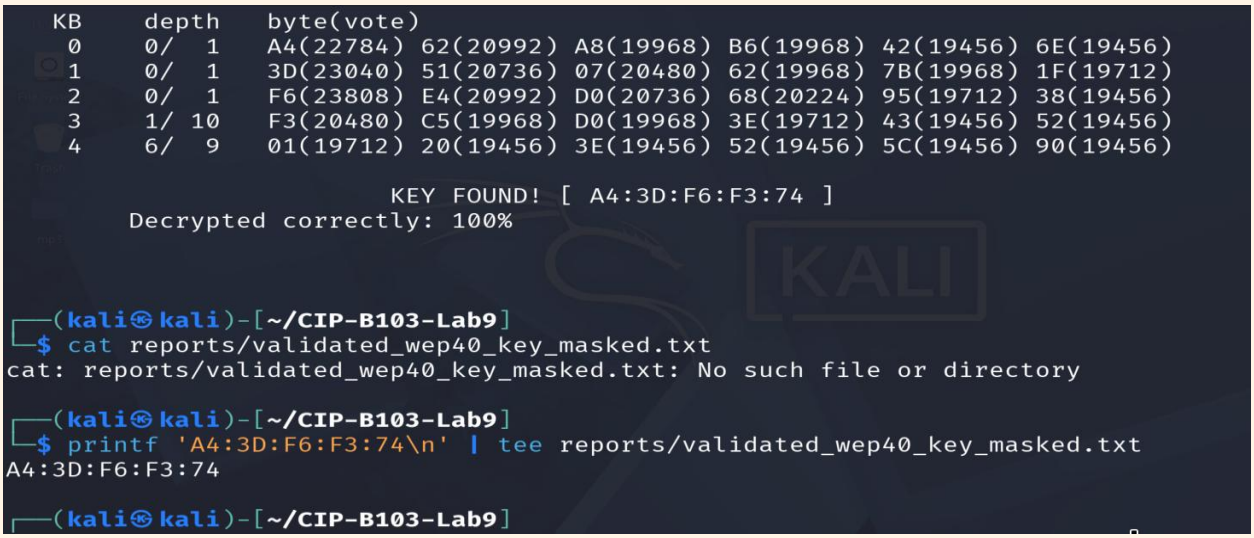
Screenshot 26: aircrack-ng launched against the capture, identifying BSSID 00:26:66:55:97:D6 (ESSID cgnetwork, WEP).



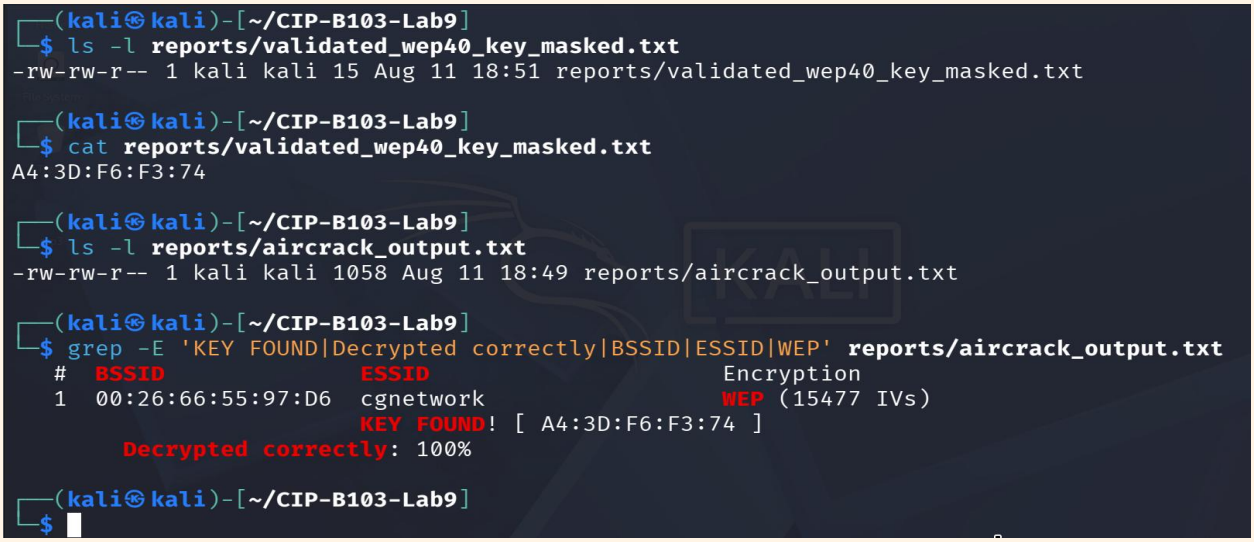
Screenshot 27: aircrack-ng PTW attack completing — KEY FOUND: A4:3D:F6:F3:74, decrypted correctly 100%.



Screenshot 28: aircrack-ng output re-confirmed via cat.



Screenshot 29: Validated WEP40 key recorded to reports/validated_wep40_key_masked.txt.



Screenshot 30: Key file and aircrack output confirmed; key line grep-summary matching the historical slide value.

11. Part D - Decrypt the Capture Offline

cd working

airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt

cd ..

find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt

sha256sum working/* | tee reports/all_working_file_hashes.txt

Identify the file created by airdecap-ng, usually with a -dec suffix. Open it in Wireshark and confirm that IP/TCP/HTTP or other higher-layer protocols are now visible.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ cd ~/CIP-B103-Lab9/working

┌──(kali㉿kali)-[~/CIP-B103-Lab9/working]
└─$ pwd
/home/kali/CIP-B103-Lab9/working

┌──(kali㉿kali)-[~/CIP-B103-Lab9/working]
└─$ airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
Total number of stations seen            10
Total number of packets read          45169
Total number of WEP data packets      15477
Total number of WPA data packets          0
Number of plaintext data packets          0
Number of decrypted WEP  packets      15477
Number of corrupted WEP  packets          0
Number of decrypted WPA  packets          0
Number of bad TKIP (WPA) packets          0
Number of bad CCMP (WPA) packets          0
```

Screenshot 31: airdecap-ng decrypting the capture with the recovered key — 15,477 WEP packets decrypted.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9/working]
└─$ cd ..

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
   526448  13904 -rw-rw-r--   1 kali     kali     14234002 Aug 11 18:18 working/file_working
   526458  12488 -rw-rw-r--   1 kali     kali     12784501 Aug 11 18:58 working/file_working-
dec
   526447  12540 -rw-rw-r--   1 kali     kali     12837660 Aug 11 18:18 working/file_working.
xz

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ sha256sum working/* | tee reports/all_working_file_hashes.txt
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04  working/file_working-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_working.xz

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$
```

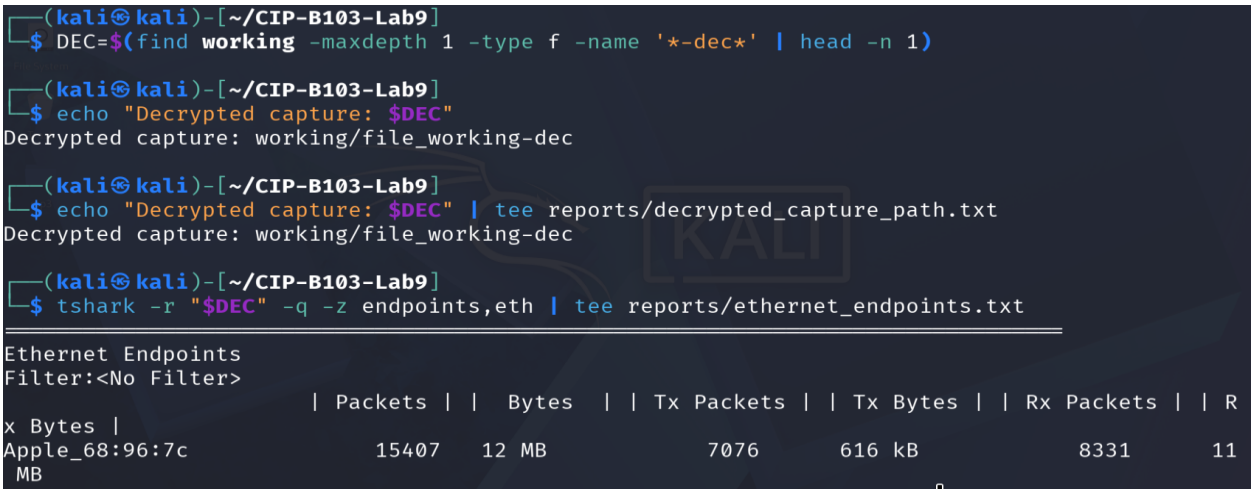
Screenshot 32: Decrypted file inventory (file_working-dec) and SHA-256 hashes of all working files.

12. Part E - Extract Endpoints and Conversations

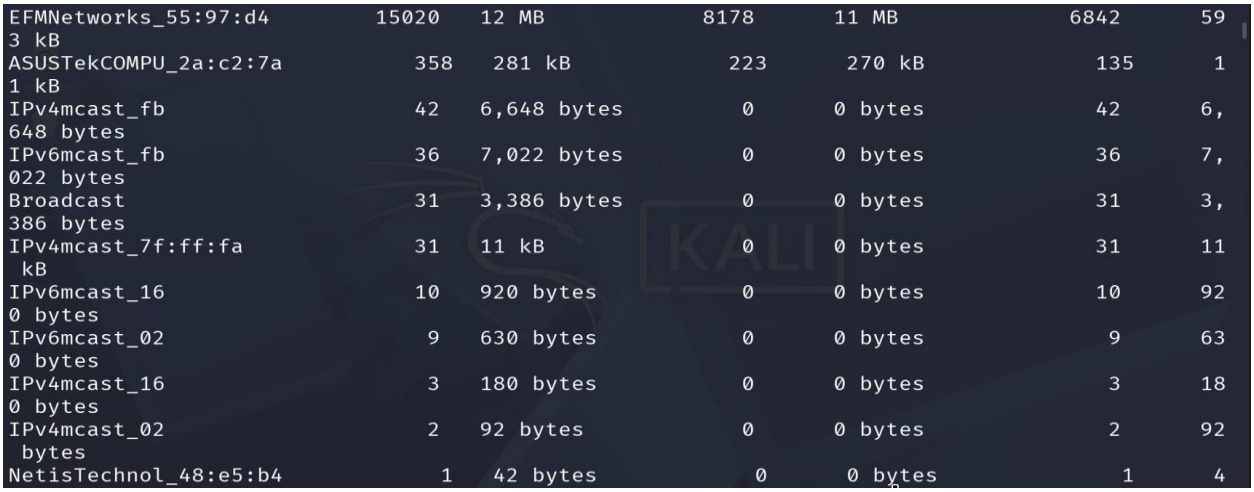
```
DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt

tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt

tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e
_ws.col.Protocol \
  | head -n 200 | tee reports/ip_mac_mapping_sample.tsv
```



Screenshot 33: DEC variable set to the decrypted capture; Ethernet endpoint statistics started.



Screenshot 34: Ethernet endpoints concluded, showing the Apple client and AP as dominant talkers.

```
2 bytes
NetisTechnol_48:a8:b5            1   42 bytes            0       0 bytes             1       4
2 bytes
08:10:77:92:7c:2f                1   42 bytes            0       0 bytes             1      42
 bytes
TendaTechnol_58:c4:ba            1   42 bytes            0       0 bytes             1       4
2 bytes
IPv6mcast_ff:2a:c2:7a            1   78 bytes            0       0 bytes             1       7
8 bytes
═══════════════════════════════════════════════════════════════════════════════

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
═══════════════════════════════════════════════════════════════════════════════
IPv4 Endpoints
Filter:<No Filter>
                      | Packets | |  Bytes  | | Tx Packets | | Tx Bytes | | Rx Packets | | R
x Bytes |
192.168.0.15              15309   12 MB           6995      608 kB           8314       11
 MB
198.90.20.111              8057   6,916 kB        4446      6,637 kB         3611       27
```

Screenshot 35: IPv4 endpoint statistics started, led by host 192.168.0.15.

```
1 kB
192.168.0.9                329   277 kB           197       265 kB           132       11
 kB
173.194.127.141            153   109 kB            80       102 kB            73       7,
364 bytes
173.194.127.55             143   89 kB             71       82 kB             72       6,
990 bytes
8.8.8.8                     81   10 kB             40       6,960 bytes       41       3,
391 bytes
199.16.158.186              66   15 kB             29       10 kB             37       5,
214 bytes
173.194.127.227             66   24 kB             26       15 kB             40       8,
455 bytes
173.194.127.229             55   22 kB             23       13 kB             32       9,
731 bytes
31.13.70.1                  51   19 kB             26       15 kB             25       3,
151 bytes
173.241.248.220             44   6,848 bytes       20       3,122 bytes       24       3,
726 bytes
198.41.208.140              44   28 kB             21       25 kB             23       2,
718 bytes
```

Screenshot 36: IPv4 endpoints continued.

```
224.0.0.251                 42   6,648 bytes        0       0 bytes           42       6,
648 bytes
173.194.127.237             36   9,526 bytes       14       6,435 bytes       22       3,
091 bytes
216.58.220.234              34   15 kB             16       12 kB             18       2,
497 bytes
192.168.0.1                 33   12 kB             33       12 kB              0       0
bytes
173.241.248.219             33   5,733 bytes       14       2,807 bytes       19       2,
926 bytes
104.74.158.12               33   21 kB             18       20 kB             15       1,
236 bytes
74.125.203.156              32   20 kB             16       17 kB             16       2,
534 bytes
239.255.255.250             31   11 kB              0       0 bytes           31       11
 kB
173.194.127.231             31   6,816 bytes       12       5,197 bytes       19       1,
619 bytes
198.41.209.137              28   16 kB             13       14 kB             15       1,
556 bytes
17.172.239.31               25   6,504 bytes       12       3,973 bytes       13       2,
```

Screenshot 37: IPv4 endpoints continued.

```
531 bytes
173.194.72.95          23   4,077 bytes    9    2,232 bytes    14    1,
845 bytes
173.194.127.122        21   12 kB          11   11 kB          10    1,
759 bytes
69.59.169.185          18   5,644 bytes    7    3,675 bytes    11    1,
969 bytes
72.246.103.16          16   3,697 bytes    6    1,732 bytes    10    1,
965 bytes
23.23.127.192          15   3,383 bytes    6    1,370 bytes    9     2,
013 bytes
173.194.127.198        14   4,480 bytes    6    2,931 bytes    8     1,
549 bytes
173.194.122.24         14   1,724 bytes    5    688 bytes      9     1,
036 bytes
23.21.133.99           13   2,314 bytes    5    880 bytes      8     1,
434 bytes
50.31.251.34           12   3,483 bytes    6    2,777 bytes    6     70
6 bytes
108.168.254.139        11   2,655 bytes    5    1,690 bytes    6     96
5 bytes
```

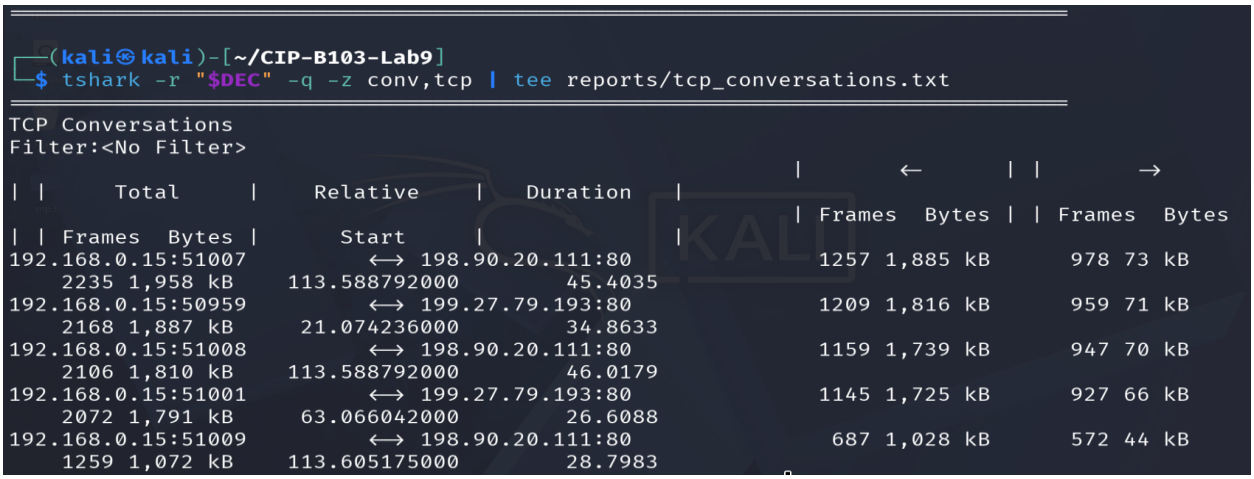
Screenshot 38: IPv4 endpoints continued.

```
108.168.254.138        11   2,333 bytes    5    1,124 bytes    6     1,
209 bytes
54.240.248.207         10   1,217 bytes    4    441 bytes      6     77
6 bytes
173.194.127.100        10   1,921 bytes    4    715 bytes      6     1,
206 bytes
54.169.42.244          10   2,054 bytes    4    1,054 bytes    6     1,
000 bytes
173.194.127.119        10   1,534 bytes    4    670 bytes      6     86
4 bytes
173.194.127.128         9   1,468 bytes    3    604 bytes      6     86
4 bytes
23.21.196.154           8   975 bytes      2    140 bytes      6     83
5 bytes
203.190.124.19          8   1,742 bytes    3    648 bytes      5     1,
094 bytes
255.255.255.255         6   2,300 bytes    0    0 bytes        6     2,
300 bytes
0.0.0.0                 5   1,710 bytes    5    1,710 bytes    0     0
bytes
173.194.127.173         4   240 bytes      2    132 bytes      2     10
```

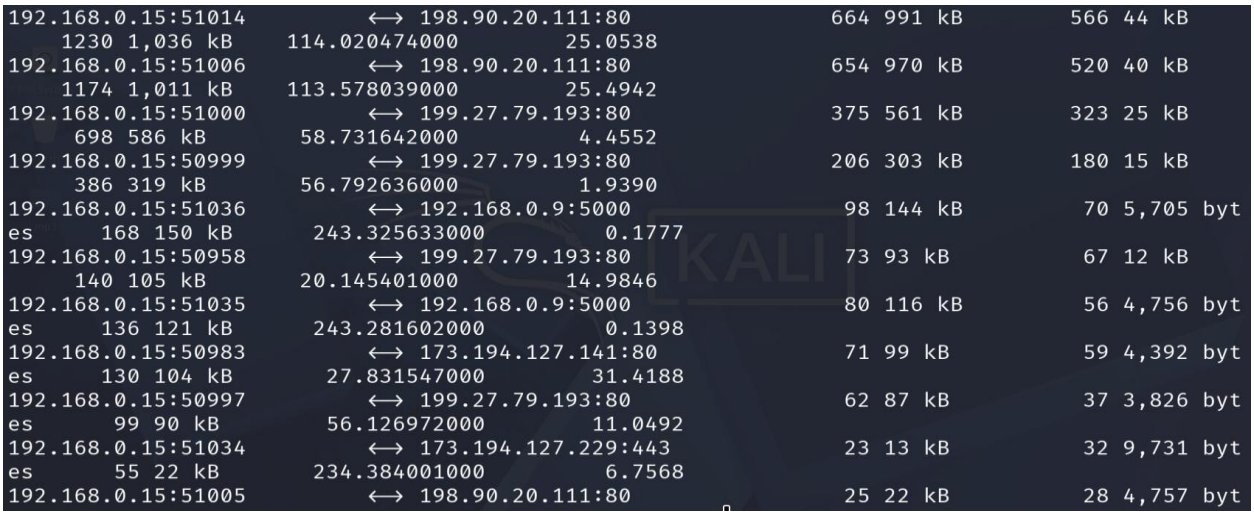
Screenshot 39: IPv4 endpoints continued.

```
8 bytes
17.172.239.127          3   206 bytes      1    74 bytes       2     13
2 bytes
173.194.122.15          3   206 bytes      1    74 bytes       2     13
2 bytes
108.168.254.140         3   198 bytes      1    66 bytes       2     13
2 bytes
173.194.127.226         3   206 bytes      1    74 bytes       2     13
2 bytes
224.0.0.22              3   180 bytes      0    0 bytes        3     18
0 bytes
224.0.0.2               2   92 bytes       0    0 bytes        2     92
 bytes
17.172.232.98           2   157 bytes      1    103 bytes      1     54
 bytes
54.251.179.109          2   132 bytes      0    0 bytes        2     13
2 bytes
173.194.122.23          2   132 bytes      0    0 bytes        2     13
2 bytes
10.0.0.1                1   60 bytes       1    60 bytes       0     0
bytes
```

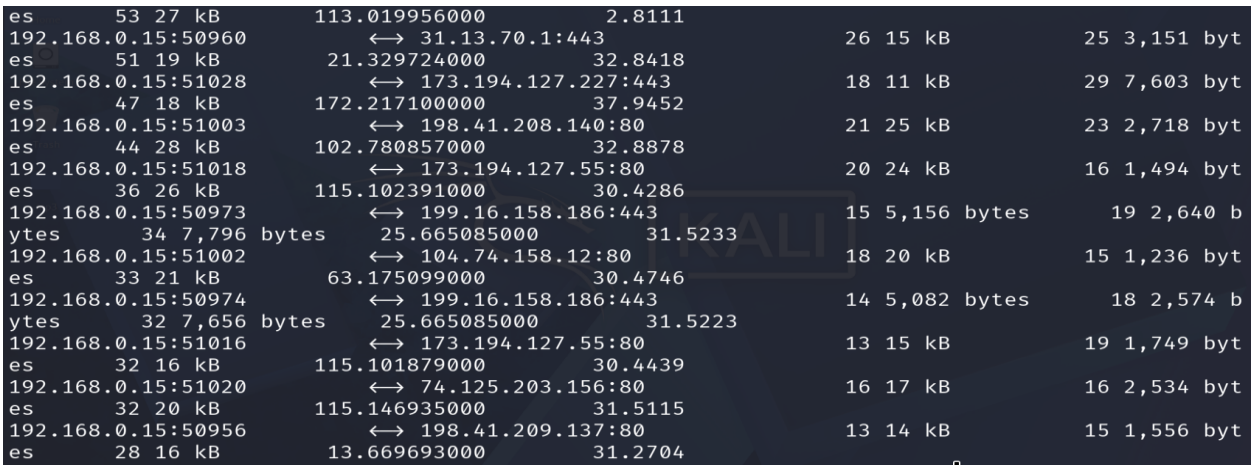
Screenshot 40: IPv4 endpoints concluded; TCP conversation statistics started.



Screenshot 41: TCP conversations — host 192.168.0.15's largest sessions to external web servers.



Screenshot 42: TCP conversations continued.



Screenshot 43: TCP conversations concluded; IP/MAC mapping field extraction started (DHCP/DNS).

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$DEC" -Y 'ip' -T fields \
-e frame.number \
-e wlan.sa \
-e wlan.da \
-e ip.src \
-e ip.dst \
-e _ws.col.Protocol \
| head -n 200 | tee reports/ip_mac_mapping_sample.tsv
7                       0.0.0.0 255.255.255.255 DHCP
9                       0.0.0.0 255.255.255.255 DHCP
11                      192.168.0.1     255.255.255.255 DHCP
20                      192.168.0.15    8.8.8.8 DNS
21                      192.168.0.15    8.8.8.8 DNS
22                      192.168.0.15    224.0.0.2       IGMPv2
23                      192.168.0.15    224.0.0.251     IGMPv2
24                      192.168.0.15    224.0.0.2       IGMPv2
25                      192.168.0.15    224.0.0.251     IGMPv2
26                      8.8.8.8 192.168.0.15    DNS
27                      8.8.8.8 192.168.0.15    DNS
```

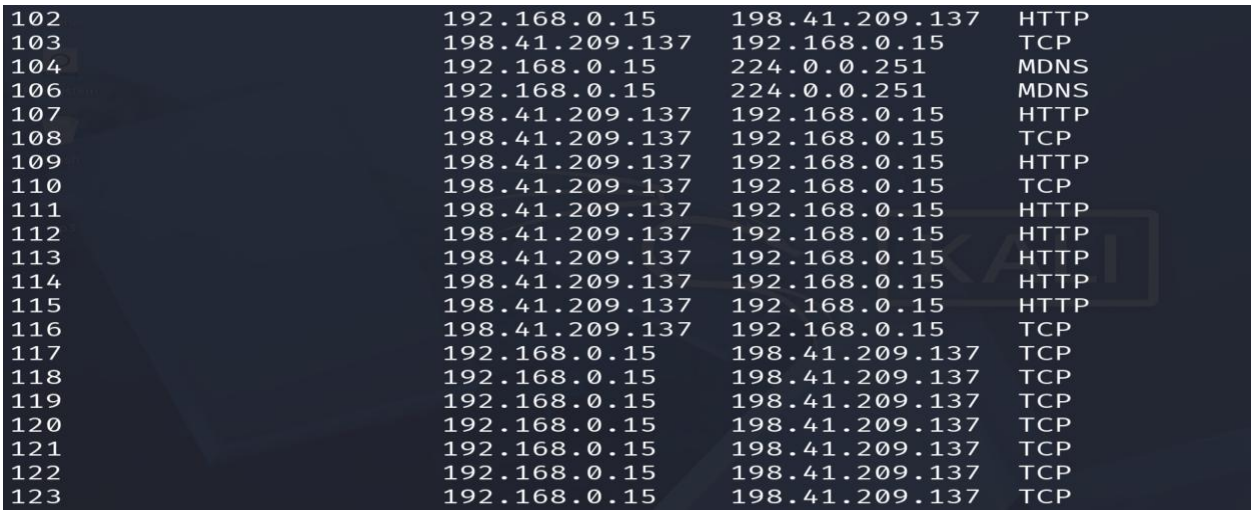
Screenshot 44: IP/MAC mapping continued — DHCP, DNS and IGMP traffic.

```
30                      192.168.0.15    17.172.239.31   TCP
31                      192.168.0.15    17.172.239.127  TCP
33                      17.172.239.31   192.168.0.15    TCP
35                      192.168.0.15    17.172.239.31   TCP
36                      192.168.0.15    17.172.239.31   TLSv1
37                      17.172.239.127  192.168.0.15    TCP
38                      192.168.0.15    17.172.239.127  TCP
42                      17.172.239.31   192.168.0.15    TCP
43                      17.172.239.31   192.168.0.15    TLSv1
44                      17.172.239.31   192.168.0.15    TLSv1
45                      17.172.239.31   192.168.0.15    TLSv1
47                      0.0.0.0 255.255.255.255 DHCP
48                      0.0.0.0 255.255.255.255 DHCP
49                      192.168.0.15    17.172.239.31   TCP
50                      192.168.0.15    17.172.239.31   TCP
51                      192.168.0.1     192.168.0.15    DHCP
52                      192.168.0.15    17.172.239.31   TLSv1
55                      17.172.239.31   192.168.0.15    TCP
56                      192.168.0.15    17.172.239.31   TLSv1
57                      17.172.239.31   192.168.0.15    TCP
58                      17.172.239.31   192.168.0.15    TLSv1
```

Screenshot 45: IP/MAC mapping continued — TCP/TLS sessions to 17.172.239.x.

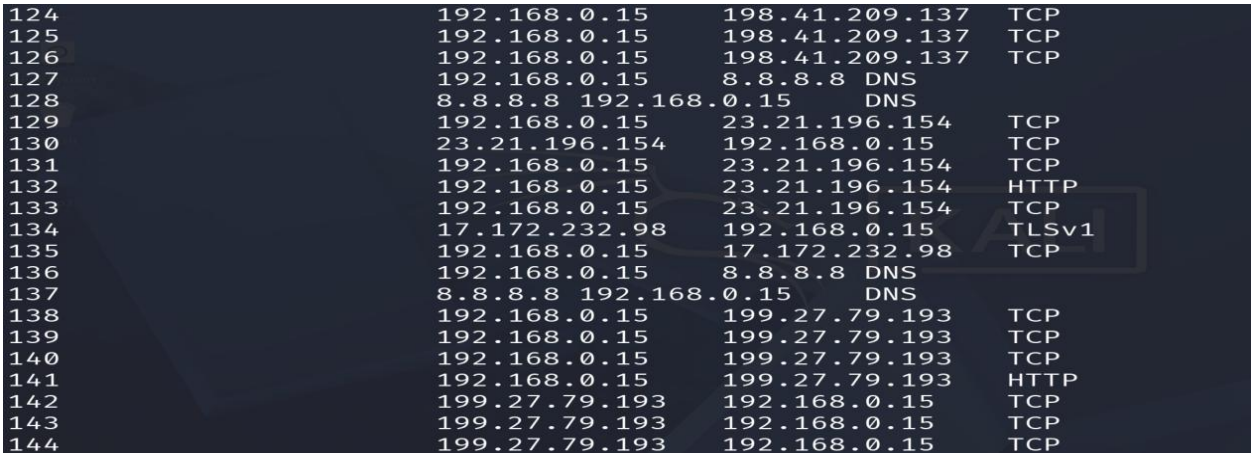
```
59                      192.168.0.15    17.172.239.31   TCP
60                      192.168.0.15    17.172.239.31   TLSv1
61                      17.172.239.31   192.168.0.15    TLSv1
65                      192.168.0.15    17.172.239.31   TCP
66                      192.168.0.15    17.172.239.31   TLSv1
69                      17.172.239.31   192.168.0.15    TCP
70                      192.168.0.15    17.172.239.31   TLSv1
74                      17.172.239.31   192.168.0.15    TCP
75                      17.172.239.31   192.168.0.15    TLSv1
78                      192.168.0.15    17.172.239.31   TCP
81                      192.168.0.15    224.0.0.251     MDNS
83                      192.168.0.15    224.0.0.251     MDNS
85                      192.168.0.15    224.0.0.251     MDNS
87                      192.168.0.15    224.0.0.251     MDNS
91                      192.168.0.15    224.0.0.251     MDNS
93                      192.168.0.15    224.0.0.251     MDNS
97                      192.168.0.15    8.8.8.8 DNS
98                      8.8.8.8 192.168.0.15    DNS
99                      192.168.0.15    198.41.209.137  TCP
100                     198.41.209.137  192.168.0.15    TCP
101                     192.168.0.15    198.41.209.137  TCP
```

Screenshot 46: IP/MAC mapping continued.



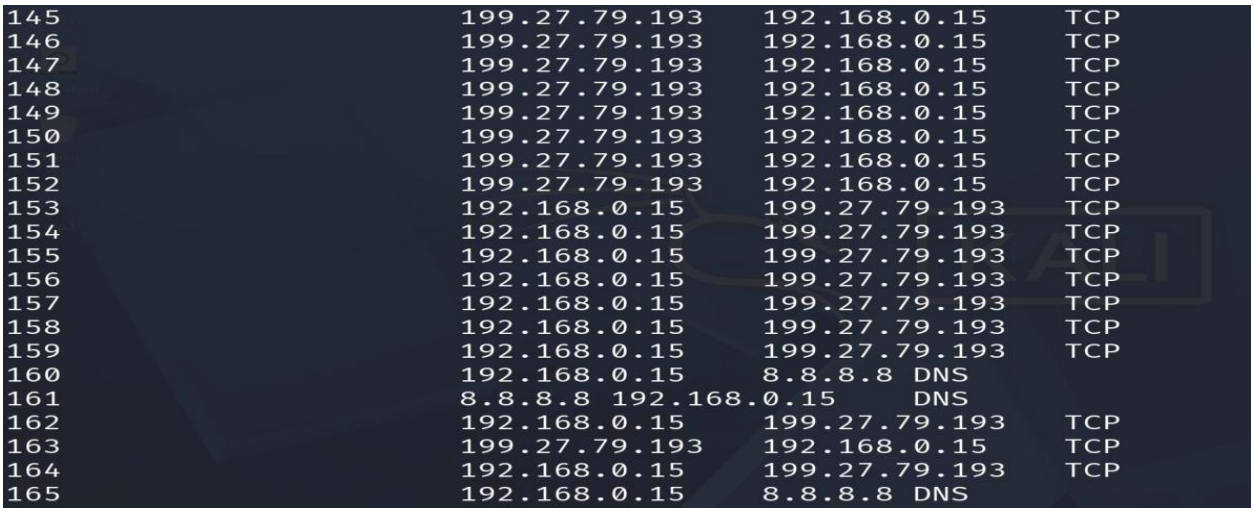
102	192.168.0.15	198.41.209.137	HTTP
103	198.41.209.137	192.168.0.15	TCP
104	192.168.0.15	224.0.0.251	MDNS
106	192.168.0.15	224.0.0.251	MDNS
107	198.41.209.137	192.168.0.15	HTTP
108	198.41.209.137	192.168.0.15	TCP
109	198.41.209.137	192.168.0.15	HTTP
110	198.41.209.137	192.168.0.15	TCP
111	198.41.209.137	192.168.0.15	HTTP
112	198.41.209.137	192.168.0.15	HTTP
113	198.41.209.137	192.168.0.15	HTTP
114	198.41.209.137	192.168.0.15	HTTP
115	198.41.209.137	192.168.0.15	HTTP
116	198.41.209.137	192.168.0.15	TCP
117	192.168.0.15	198.41.209.137	TCP
118	192.168.0.15	198.41.209.137	TCP
119	192.168.0.15	198.41.209.137	TCP
120	192.168.0.15	198.41.209.137	TCP
121	192.168.0.15	198.41.209.137	TCP
122	192.168.0.15	198.41.209.137	TCP
123	192.168.0.15	198.41.209.137	TCP

Screenshot 47: IP/MAC mapping continued, including the first HTTP frames.



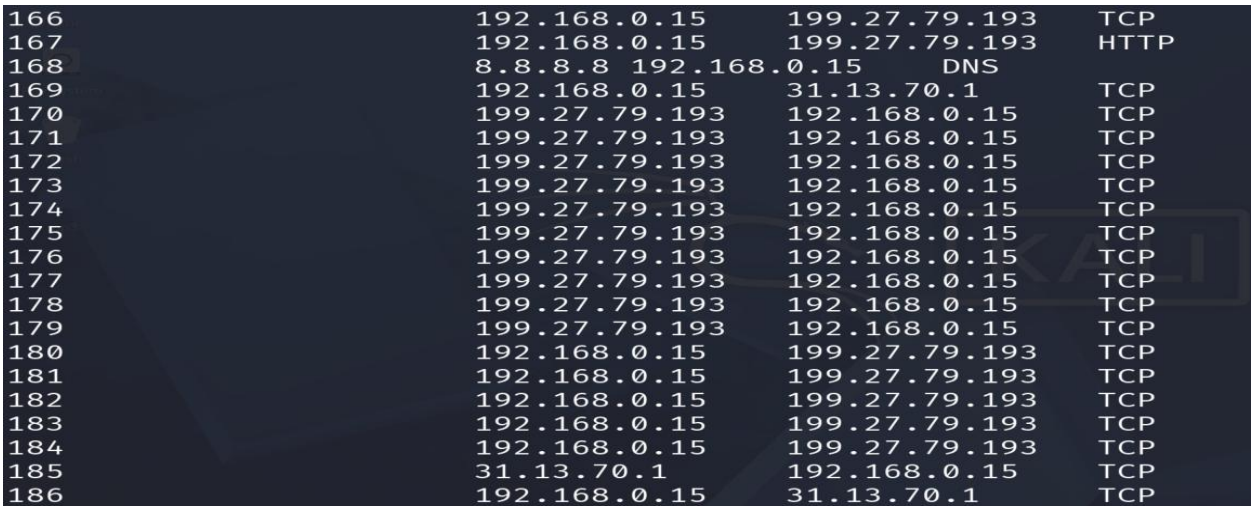
124	192.168.0.15	198.41.209.137	TCP
125	192.168.0.15	198.41.209.137	TCP
126	192.168.0.15	198.41.209.137	TCP
127	192.168.0.15	8.8.8.8	DNS
128	8.8.8.8	192.168.0.15	DNS
129	192.168.0.15	23.21.196.154	TCP
130	23.21.196.154	192.168.0.15	TCP
131	192.168.0.15	23.21.196.154	TCP
132	192.168.0.15	23.21.196.154	HTTP
133	192.168.0.15	23.21.196.154	TCP
134	17.172.232.98	192.168.0.15	TLSv1
135	192.168.0.15	17.172.232.98	TCP
136	192.168.0.15	8.8.8.8	DNS
137	8.8.8.8	192.168.0.15	DNS
138	192.168.0.15	199.27.79.193	TCP
139	192.168.0.15	199.27.79.193	TCP
140	192.168.0.15	199.27.79.193	TCP
141	192.168.0.15	199.27.79.193	HTTP
142	199.27.79.193	192.168.0.15	TCP
143	199.27.79.193	192.168.0.15	TCP
144	199.27.79.193	192.168.0.15	TCP

Screenshot 48: IP/MAC mapping continued.



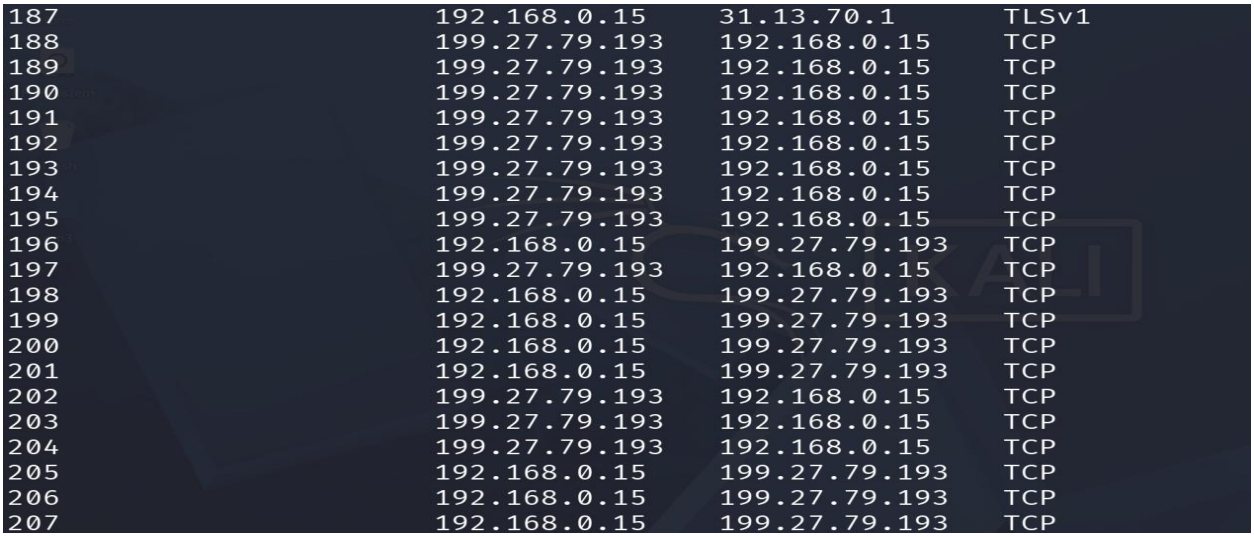
145	199.27.79.193	192.168.0.15	TCP
146	199.27.79.193	192.168.0.15	TCP
147	199.27.79.193	192.168.0.15	TCP
148	199.27.79.193	192.168.0.15	TCP
149	199.27.79.193	192.168.0.15	TCP
150	199.27.79.193	192.168.0.15	TCP
151	199.27.79.193	192.168.0.15	TCP
152	199.27.79.193	192.168.0.15	TCP
153	192.168.0.15	199.27.79.193	TCP
154	192.168.0.15	199.27.79.193	TCP
155	192.168.0.15	199.27.79.193	TCP
156	192.168.0.15	199.27.79.193	TCP
157	192.168.0.15	199.27.79.193	TCP
158	192.168.0.15	199.27.79.193	TCP
159	192.168.0.15	199.27.79.193	TCP
160	192.168.0.15	8.8.8.8	DNS
161	8.8.8.8	192.168.0.15	DNS
162	192.168.0.15	199.27.79.193	TCP
163	199.27.79.193	192.168.0.15	TCP
164	192.168.0.15	199.27.79.193	TCP
165	192.168.0.15	8.8.8.8	DNS

Screenshot 49: IP/MAC mapping continued.



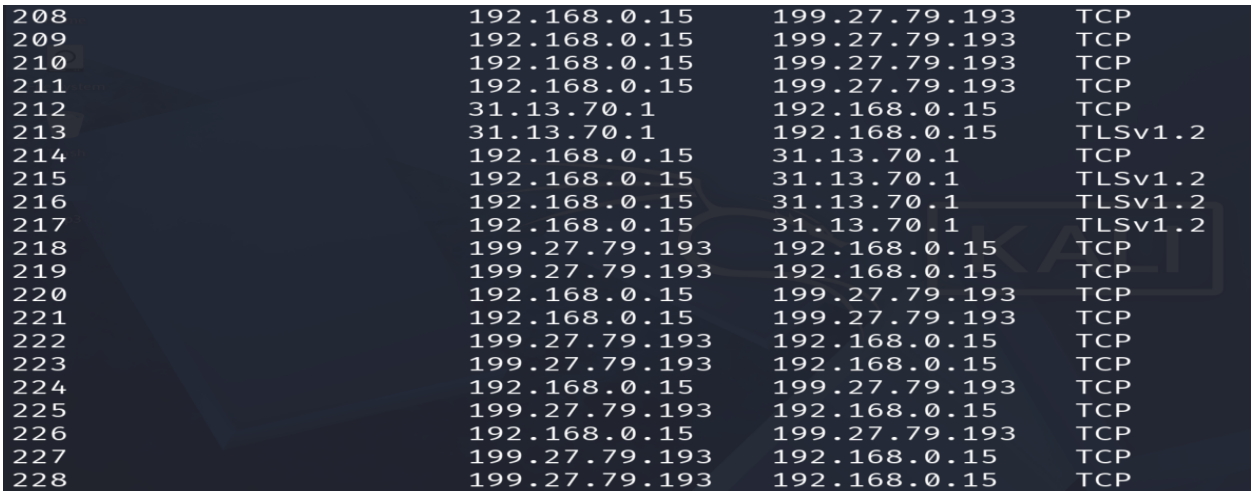
166	192.168.0.15	199.27.79.193	TCP
167	192.168.0.15	199.27.79.193	HTTP
168	8.8.8.8	192.168.0.15	DNS
169	192.168.0.15	31.13.70.1	TCP
170	199.27.79.193	192.168.0.15	TCP
171	199.27.79.193	192.168.0.15	TCP
172	199.27.79.193	192.168.0.15	TCP
173	199.27.79.193	192.168.0.15	TCP
174	199.27.79.193	192.168.0.15	TCP
175	199.27.79.193	192.168.0.15	TCP
176	199.27.79.193	192.168.0.15	TCP
177	199.27.79.193	192.168.0.15	TCP
178	199.27.79.193	192.168.0.15	TCP
179	199.27.79.193	192.168.0.15	TCP
180	192.168.0.15	199.27.79.193	TCP
181	192.168.0.15	199.27.79.193	TCP
182	192.168.0.15	199.27.79.193	TCP
183	192.168.0.15	199.27.79.193	TCP
184	192.168.0.15	199.27.79.193	TCP
185	31.13.70.1	192.168.0.15	TCP
186	192.168.0.15	31.13.70.1	TCP

Screenshot 50: IP/MAC mapping continued.



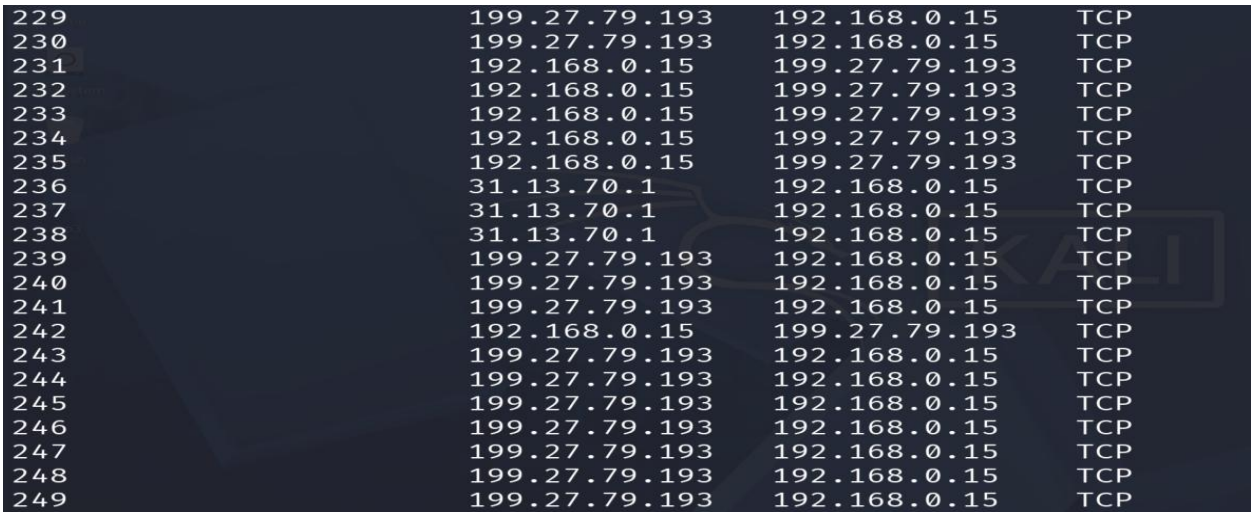
187	192.168.0.15	31.13.70.1	TLSv1
188	199.27.79.193	192.168.0.15	TCP
189	199.27.79.193	192.168.0.15	TCP
190	199.27.79.193	192.168.0.15	TCP
191	199.27.79.193	192.168.0.15	TCP
192	199.27.79.193	192.168.0.15	TCP
193	199.27.79.193	192.168.0.15	TCP
194	199.27.79.193	192.168.0.15	TCP
195	199.27.79.193	192.168.0.15	TCP
196	192.168.0.15	199.27.79.193	TCP
197	199.27.79.193	192.168.0.15	TCP
198	192.168.0.15	199.27.79.193	TCP
199	192.168.0.15	199.27.79.193	TCP
200	192.168.0.15	199.27.79.193	TCP
201	192.168.0.15	199.27.79.193	TCP
202	199.27.79.193	192.168.0.15	TCP
203	199.27.79.193	192.168.0.15	TCP
204	199.27.79.193	192.168.0.15	TCP
205	192.168.0.15	199.27.79.193	TCP
206	192.168.0.15	199.27.79.193	TCP
207	192.168.0.15	199.27.79.193	TCP

Screenshot 51: IP/MAC mapping continued.



208	192.168.0.15	199.27.79.193	TCP
209	192.168.0.15	199.27.79.193	TCP
210	192.168.0.15	199.27.79.193	TCP
211	192.168.0.15	199.27.79.193	TCP
212	31.13.70.1	192.168.0.15	TCP
213	31.13.70.1	192.168.0.15	TLSv1.2
214	192.168.0.15	31.13.70.1	TCP
215	192.168.0.15	31.13.70.1	TLSv1.2
216	192.168.0.15	31.13.70.1	TLSv1.2
217	192.168.0.15	31.13.70.1	TLSv1.2
218	199.27.79.193	192.168.0.15	TCP
219	199.27.79.193	192.168.0.15	TCP
220	192.168.0.15	199.27.79.193	TCP
221	192.168.0.15	199.27.79.193	TCP
222	199.27.79.193	192.168.0.15	TCP
223	199.27.79.193	192.168.0.15	TCP
224	192.168.0.15	199.27.79.193	TCP
225	199.27.79.193	192.168.0.15	TCP
226	192.168.0.15	199.27.79.193	TCP
227	199.27.79.193	192.168.0.15	TCP
228	199.27.79.193	192.168.0.15	TCP

Screenshot 52: IP/MAC mapping continued.

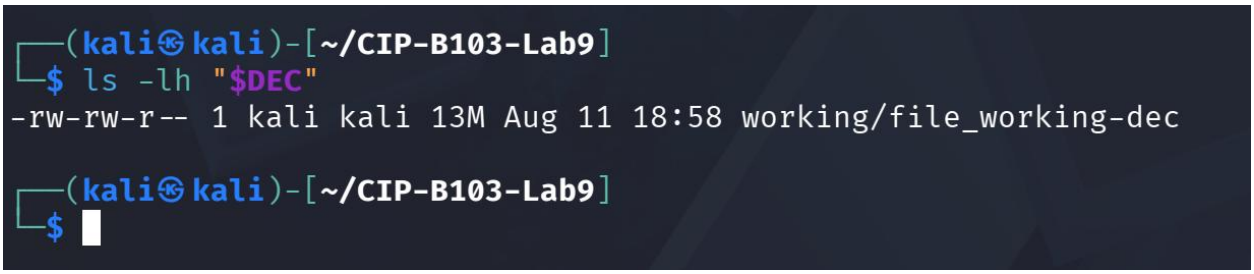


Screenshot 53: IP/MAC mapping concluded (200-frame sample).

13. Part F - Extract Images, HTML and Other Objects

5. First try Wireshark File > Export Objects for HTTP or other recognized protocols.
6. Use tshark object export for HTTP where available.
7. Use foremost as a carving method and document that carved files may include false positives or lack original filenames.
8. Open recovered objects only inside the lab VM and hash them before viewing.

```
mkdir -p exported/http_objects exported/foremost
# HTTP object export if HTTP is present
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
# Generic carving
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
```



Screenshot 54: Decrypted capture size confirmed; export directories created.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ mkdir -p exported/http_objects exported/foremost

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_expo
rt_log.txt
    1   0.000000 EFMNetworks_55:97:d4 → Broadcast    ARP 42 Who has 192.168.0.15? Tell 192.16
8.0.1
    2   0.943674 Apple_68:96:7c → NetisTechnol_48:e5:b4 ARP 42 Who has 192.168.1.1? Tell 192.
168.1.41
    3   0.944186 Apple_68:96:7c → NetisTechnol_48:a8:b5 ARP 42 Who has 192.168.0.1? Tell 192.
168.0.15
    4   0.944186 Apple_68:96:7c → 08:10:77:92:7c:2f ARP 42 Who has 192.168.1.1? Tell 192.168.
1.12
    5   0.944186 Apple_68:96:7c → EFMNetworks_55:97:d4 ARP 42 Who has 192.168.0.1? Tell 192.1
68.0.15
    6   0.944186 Apple_68:96:7c → TendaTechnol_58:c4:ba ARP 42 Who has 192.168.10.1? Tell 192
.168.10.113
    7   0.944185       0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request  - Transaction ID 0×21e
6be1c
```

Screenshot 55: tshark HTTP object export started, showing early ARP/DHCP frames.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported/http_objects -type f -ls
   526566      4 -rw-r--r--    1 kali     kali         2671 Aug 11 19:26 exported/http_objects
/logo.png
   526535      4 -rw-r--r--    1 kali     kali          129 Aug 11 19:26 exported/http_objects
/&i=19612&llnocookies=undefined
   526473    180 -rw-r--r--    1 kali     kali       181022 Aug 11 19:26 exported/http_objects
/hit.json
   526556      4 -rw-r--r--    1 kali     kali         3928 Aug 11 19:26 exported/http_objects
/rss.png
   526538     16 -rw-r--r--    1 kali     kali        12768 Aug 11 19:26 exported/http_objects
/qpy-UiLNKP-VfOdbcs6r6-vvDin1pK8aKteLpeZ5c0A.woff
   526493      4 -rw-r--r--    1 kali     kali          166 Aug 11 19:26 exported/http_objects
/adc-i-00aecd.png
   526523      8 -rw-r--r--    1 kali     kali         4862 Aug 11 19:26 exported/http_objects
/superfish.min.js
   526553      4 -rw-r--r--    1 kali     kali         1226 Aug 11 19:26 exported/http_objects
/category-strip.png
   526530     12 -rw-r--r--    1 kali     kali        12076 Aug 11 19:26 exported/http_objects
```

Screenshot 56: HTTP objects recovered — logo.png, tracking pixel, hit.json, rss.png, a web font and more.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
Processing: working/file_working-dec
|*|

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported/foremost -type f -ls
   526613      4 -rw-rw-r--    1 kali     kali         1604 Aug 11 19:27 exported/foremost/aud
it.txt
   526614    260 -rw-rw-r--    1 kali     kali       264865 Aug 11 19:27 exported/foremost/jpg
/00000104.jpg
   526615     88 -rw-rw-r--    1 kali     kali        89458 Aug 11 19:27 exported/foremost/jpg
/00004538.jpg
   526619      4 -rw-rw-r--    1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00003653.gif
   526618      4 -rw-rw-r--    1 kali     kali           42 Aug 11 19:27 exported/foremost/gif
/00002450.gif
   526617      4 -rw-rw-r--    1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00001260.gif
```

Screenshot 57: foremost carving run against the decrypted capture, recovering JPEG and GIF files.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported/foremost -maxdepth 2 -type d -print
exported/foremost
exported/foremost/jpg
exported/foremost/gif
exported/foremost/png
exported/foremost/htm

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
exported/foremost/audit.txt: ASCII text
exported/foremost/jpg/00000104.jpg: JPEG image data, JFIF standard 1.01, aspect ratio, densit
y 1×1, segment length 16, baseline, precision 8, 2492×2492, components 3
exported/foremost/jpg/00004538.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI), de
nsity 72×72, segment length 16, comment: "Lavc55.69.100", baseline, precision 8, 718×404, com
ponents 3
exported/foremost/gif/00003653.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00002450.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00001260.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00000406.gif: GIF image data, version 89a, 1 x 1
```

Screenshot 58: foremost output directories confirmed (jpg, gif, png, htm).

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
d54f38f5df345950c96276c918d553695e4bd504c3c4c808324b971b576f7265  exported/foremost/audit.txt
3f9bed324e04c1735b498dab0f138673e20591a5b39201e731a91dd180929ba6  exported/foremost/jpg/00000
104.jpg
f891fa7ef7652a26a54638dd7b0aebcb60676fd72ec42a2a636523a6371297dc  exported/foremost/jpg/00004
538.jpg
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00003
653.gif
ef1955ae757c8b966c83248350331bd3a30f658ced11f387f8ebf05ab3368629  exported/foremost/gif/00002
450.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00001
260.gif
a0d3a0aff7dc3bf32d2176fc3dcda6e7aba2867c4f4d1f7af6355d2cfc6c44f8  exported/foremost/gif/00000
406.gif
53f7e56783c414746a9fc3612637509560424617e3135248640eac142d6e0f94  exported/foremost/gif/00010
456.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00003
719.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00010
```

Screenshot 59: file-type identification and SHA-256 hashing of all carved objects.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported -type f -ls
   526613      4 -rw-rw-r--   1 kali     kali         1604 Aug 11 19:27 exported/foremost/aud
it.txt
   526614    260 -rw-rw-r--   1 kali     kali       264865 Aug 11 19:27 exported/foremost/jpg
/00000104.jpg
   526615     88 -rw-rw-r--   1 kali     kali        89458 Aug 11 19:27 exported/foremost/jpg
/00004538.jpg
   526619      4 -rw-rw-r--   1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00003653.gif
   526618      4 -rw-rw-r--   1 kali     kali           42 Aug 11 19:27 exported/foremost/gif
/00002450.gif
   526617      4 -rw-rw-r--   1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00001260.gif
   526616      4 -rw-rw-r--   1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00000406.gif
   526621      4 -rw-rw-r--   1 kali     kali           49 Aug 11 19:27 exported/foremost/gif
/00010456.gif
   526620      4 -rw-rw-r--   1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00003719.gif
```

Screenshot 60: Carved file inventory reconfirmed via find.

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ ls -lh reports/http_export_log.txt \
      reports/foremost_log.txt \
      reports/exported_file_types.txt \
      reports/exported_file_hashes.txt
-rw-rw-r-- 1 kali kali  17K Aug 11 19:29 reports/exported_file_hashes.txt
-rw-rw-r-- 1 kali kali  14K Aug 11 19:29 reports/exported_file_types.txt
-rw-rw-r-- 1 kali kali    0 Aug 11 19:27 reports/foremost_log.txt
-rw-rw-r-- 1 kali kali 2.3M Aug 11 19:26 reports/http_export_log.txt

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported -type f -ls
cat reports/exported_file_types.txt
cat reports/exported_file_hashes.txt
   526613      4 -rw-rw-r--   1 kali     kali         1604 Aug 11 19:27 exported/foremost/aud
it.txt
   526614    260 -rw-rw-r--   1 kali     kali       264865 Aug 11 19:27 exported/foremost/jpg
/00000104.jpg
   526615     88 -rw-rw-r--   1 kali     kali        89458 Aug 11 19:27 exported/foremost/jpg
/00004538.jpg
```

Screenshot 61: Final report file sizes confirmed for the object-extraction logs and hash lists.

14. Part G - Optional Historical Passphrase Challenge

Optional and Instructor-Assessed

The slide discusses deriving a seed and cracking the CTF passphrase using supplied challenge

information. This is not required for the core WEP decryption outcome unless assigned. Do not adapt the code to live networks.

- Differentiate the 40-bit WEP key from the human-readable router passphrase.
- Record the known key, derived seed, stated character constraints and challenge hash.
- Run only the instructor-provided findseed.py or crackWEP40 script against the historical dataset.
- Document the recovered passphrase in a protected appendix, not in public screenshots.

15. WEP Security Analysis

- WEP uses RC4 with a short 24-bit IV, allowing IV reuse in busy networks.
- The same keystream reuse leaks relationships between plaintexts and enables statistical key recovery.
- WEP provides weak integrity and no modern protection against replay or key-recovery attacks.
- Recommended replacement is WPA2-AES/CCMP or WPA3 with strong authentication and protected management frames where supported.
- Legacy WEP should be removed rather than strengthened with longer passphrases.

16. Required Findings Worksheet

Question	Finding
Capture format and duration	pcap, IEEE 802.11 Wireless LAN encapsulation, microsecond timestamps; 45,169 packets over 272.969795 s (2015-03-06 02:51:54.364525 to 02:56:27.334320)
BSSID	00:26:66:55:97:D6 (ESSID ‘cgnetwork’, per aircrack-ng)
Primary station MAC addresses	f0:f6:1c:68:96:7c (Apple client, host IP 192.168.0.15); 00:26:66:55:97:d4 (AP/router Ethernet-side MAC, EFMNetworks, DHCP server 192.168.0.1)
Protected frame count	15,477 WEP data packets (confirmed independently by both airdecap-ng and aircrack-ng's IV count)
Repeated IV evidence	reports/repeated_iv_summary.txt shows IVs such as 0x42e8eb and 0x28d3eb each recurring 7 times within just the first 200-frame sample; broad reuse across all 15,477 IVs is what allowed the PTW attack to succeed

Question	Finding
Recovered/validated WEP40 key	A4:3D:F6:F3:74 — independently recovered via aircrack-ng's PTW attack (Tested 83 keys, 100% decrypted correctly), matching the historical slide-provided key
Decrypted capture filename/hash	working/file_working-dec, SHA-256 167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04
Host IP and MAC	192.168.0.15, f0:f6:1c:68:96:7c (Apple_68:96:7c) — the dominant traffic-generating client
Server IP and MAC	Multiple external HTTP/HTTPS servers (e.g. 198.90.20.111, 199.27.79.193, 198.41.209.137); their link-layer MACs are not directly observable since all external traffic transits the AP/router at 00:26:66:55:97:d4 (192.168.0.1) as the local next-hop
Key protocols observed	DHCP, ARP, DNS, IGMPv2, MDNS, TCP, HTTP, TLSv1/TLSv1.2
Recovered images/HTML	HTTP object export: logo.png, rss.png, category-strip.png, adc-i-0Qaecd.png, a WOFF web font, superfish.min.js and hit.json. foremost carving additionally recovered JPEGs (00000104.jpg 2492x2492; 00004538.jpg 718x404), several small GIFs, and an htm/ directory of carved HTML.
Optional router passphrase	Protected appendix only
Security conclusion	WEP40's 24-bit IV space is far too small for real traffic volumes, producing extensive IV reuse that let aircrack-ng's PTW attack recover the full key from 15,477 IVs with 100% decryption accuracy in well under a second — a structural RC4/WEP flaw rather than a weak-password issue. WEP should be retired in favor of WPA2-AES/CCMP or WPA3 rather than strengthened.

Required Screenshots Checklist

☐ Compressed evidence file and hash.

☐ Decompressed capture identification.

☐ Protocol hierarchy.

☐ 802.11 frame categories.

☐ Protected WEP frames and IV evidence.

☐ Aircrack-ng result.

☐ airdecap-ng decryption result.

☐ Decrypted IP/MAC endpoint inventory.

☐ Exported image or HTML objects.

☐ Hashes of decrypted capture and recovered files.

Student Report Template

9. Executive summary and authorization.
10. Evidence preparation and hashes.
11. 802.11/WEP identification.
12. Key-recovery or validation procedure.
13. Offline decryption.
14. Endpoint and conversation analysis.
15. Object extraction.
16. Optional passphrase challenge.
17. Security weaknesses and recommendations.
18. Appendix.

Submission Requirements

- One PDF report named CIP-B103-Lab9_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Evidence handling	10	Historical dataset preserved and hashed.
802.11/WEP analysis	20	Correct frame types, protected frames, BSSID and IV explanation.
Key recovery/validation	15	Correct Aircrack-ng procedure and key documentation.
Decryption	20	Successful airdecap-ng output and verification.
Endpoint/object forensics	25	IP/MAC findings and recovered files with hashes.
Report quality	10	Clear security conclusion and screenshots.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
WEP40 composition	24-bit IV plus 40-bit secret key.
Slide-provided key	A4:3D:F6:F3:74.
airdecap format	Enter key as A43DF6F374.
Primary weakness	IV reuse and RC4-related key-recovery weaknesses.
Post-decryption evidence	Higher-layer protocols, endpoints and objects become analyzable.
Modern recommendation	WPA2-AES/CCMP or WPA3; retire WEP.

Command Reference Summary

Command	Purpose
aircrack-ng capture	Recover or validate wireless key.
airdecap-ng -w KEY capture	Decrypt WEP traffic.
wlan.fc.protected==1	Filter protected 802.11 frames.
wlan.wep.iv	WEP initialization vector field.
tshark --export-objects	Extract protocol objects.
foremost -i FILE -o DIR	Carve files from evidence.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.